

RES.6-012 Introduction to Probability - Lecture 13.4, 15
Textbook Problem Sets - Sections 4.3 - 4.6 ; 8.1 - 8.2

22) $F_2 \rightarrow$ Fortune.

Bet: $F_0(2p-1)$

R1 outcome: $F_1 = F_0 + F_0(2p-1)p - F_0(2p-1)(1-p)$

$$= F_0 + F_0(2p-1)p - F_0(2p)(1-p) + F_0(1-p)$$

$$= F_0 + 2pF_0 - F_0p - 2pF_0(1-p) + F_0 - F_0p$$

$$= F_0(1 + (2p-1)p - (2p-1)(1-p))$$

$$= F_0(1 + (2p-1)(p - (1-p)))$$

$$= F_0(1 + (2p-1)^2) \quad \textcircled{1}$$

$\hookrightarrow$ Multiplicative not additive

$$F_1 \rightarrow F_0 \left(\dots \right)^2$$

Let X_k be $F_k \rightarrow$ Fortune k th round.

$$E[F_{k+1} | F_k] = \left[(1 + (2p-1)^2) \right] F_k$$

Law of Iterated Expectations: $E[E[F_{k+1} | F_k]] = E[F_{k+1}]$

$$E[F_{k+1}] = (1 + (2p-1)^2) E[F_k]$$

$$\in (1 + (2p-1)^2)^n F_0$$

$$\therefore E[F_n] = \left[1 + (2p-1)^2 \right]^n F_0$$

23) Nat arrive 9pm
 Pat arrive uniform $[8\text{pm}, 10\text{pm}]$
 $X \rightarrow$ time in hours between 8pm & Pat arriving $[0, 2]$.

If: Pat arrive before 9pm ($x \leq 1$) then date 3hrs
 after 9pm ($x > 1$) then date
 uniform $[0, 3-x]$

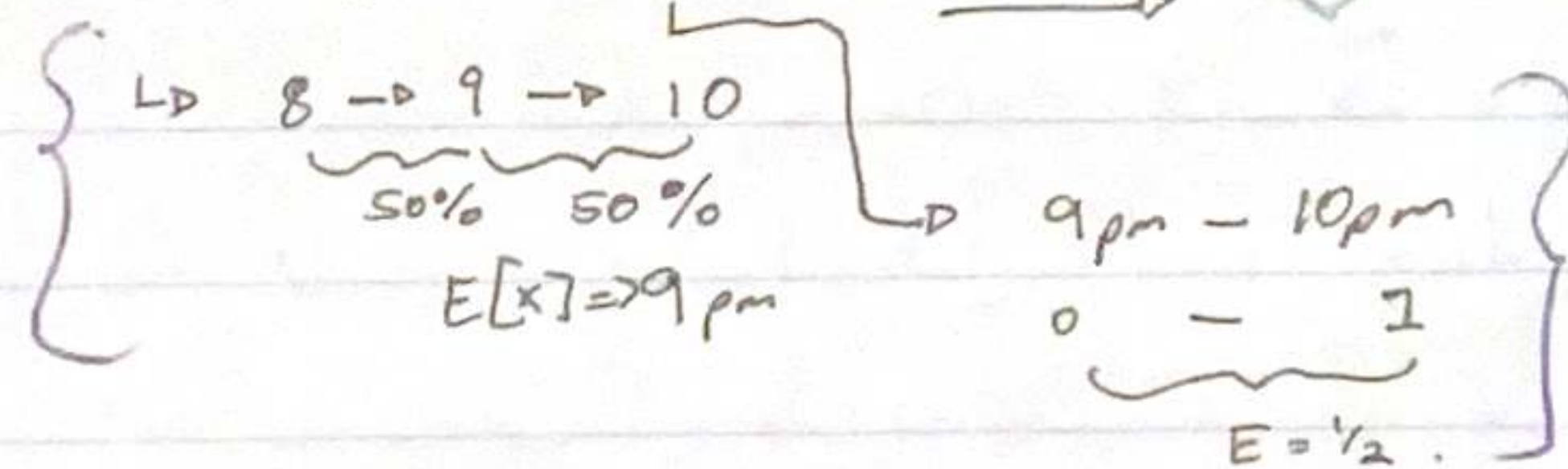
Nat end relationship If: Pat late $\geq 45\text{min}$ twice
 $X \geq 0.45$ twice (not consecutive)

a) $E[X] = ?$ Uniform $\rightarrow [0, 2]$ Let W be hours Nat waits
 $E[X] = \frac{0+2}{2} = 1$ hour

$$\hookrightarrow E[X] = P(0 \leq x \leq 1)E[W | 0 \leq x \leq 1] + P(x > 1)E[W | x > 1]$$

$$E[W] = P(x > 1)E[W | x > 1]$$

$$= \left(\frac{1}{2}\right) \left(\frac{0+1}{2}\right) = \frac{1}{4}$$



$$b) E[D] = P(0 \leq x \leq 1)(3) + P(x > 1)(x - 3)$$

$$= P E[D | 0 \leq x \leq 1] + P E[D | x > 1]$$

$$= P(0 \leq x \leq 1)(3) + P(x > 1)(E[X] - 3)$$

$$= \left(\frac{1}{2}\right)(3) + \left(\frac{1}{2}\right)\left(\left(\frac{1+2}{2}\right) - 3\right)$$

$$= 1.5 + \left(\frac{1}{2}\right)(1.5 - 3)$$

$$= 1.5 - 0.75$$

$$= 0.75 \text{ hours}$$

$$0.75 = \frac{3}{4} = \frac{3}{2} + \frac{1}{2} \left(\frac{3}{2} - \frac{3}{4} \right)$$

$$= \frac{3}{2} + \frac{1}{2} \left(\frac{3}{4} \right)$$

$$= \frac{3}{2} + \frac{3}{8}$$

$$= \frac{12}{8} + \frac{3}{8} = \frac{15}{8}$$

$$E[D | x > 1] = P(x > 1)E\left[\frac{3-x}{2} | x > 1\right]$$

$$\Rightarrow \frac{1}{2} \left(\frac{3-x}{2} \right)$$

$$\text{When } x > 1 \text{ } \circ \text{ Uniform } [0, 3-x]$$

$$E[\text{Pat Late}] = \frac{(3-x) - (0)}{2}$$

$$E[D | \text{Pat Late}] = \frac{(3-x) - (0)}{2}$$

$$\Rightarrow \frac{3}{2} - \frac{E[X | x > 1]}{2} \rightarrow [1, 2]$$

$$\Rightarrow \frac{3}{2} - \frac{(3/2)}{2}$$

$$\Rightarrow E\left[\frac{3-x}{2} | x > 1\right] = \frac{3}{2} - E[X | x > 1]$$

43) 4 lights

$$P(\text{Red Light}) = P(R) = 0.5$$

Waiting: Normal(1 min, 0.5 min)

$X \rightarrow$ total waiting time at red lights

$$a) f_x = \sum_{i=0}^3 P(R) \cdot f_{x|L}(x)$$

$$= \sum_{i=0}^4 \left(\frac{1}{2}\right) \cdot E[X_i | L=R]$$

$$= \sum_{i=0}^4 \left(\frac{1}{2}\right) \cdot (1)$$

$$P(X > 4) = \sum_{k=0}^4 P(k \text{ lights are red}) \cdot P(X > 4 | k \text{ lights red})$$

$$= \sum_{k=0}^4 (0.5)^k (1 - P(X \leq 4 | k \text{ lights red}))$$

$$= \sum_{k=0}^4 (0.5)^k \left(1 - Z\left(\frac{(4)-(1)}{(\frac{1}{2})}\right)\right)$$

$$= \sum_{k=0}^4 \left(\frac{1}{2}\right)^k \left[1 - \right]$$

$$a) p_x(x) = \sum_{i=0}^4 P(A_i) \cdot p_{x|A_i}(x) \quad \left(\begin{array}{l} \text{Red Lights } P=1/2 \ n \leq 4 \\ \text{Binomial} \end{array} \right)$$

$$= \sum_{i=0}^4 P(k \text{ Red Lights}) \cdot p_{x|kR}(x)$$

$$= \sum_{i=0}^4 \left[\binom{4}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{4-k} \right] p_{x|kR}(x)$$

$$\left[\begin{array}{l} \text{Normal}(1, (\frac{1}{2})^2) \\ \rightarrow \text{Normal}(k, k/4) \end{array} \right]$$

$$p_x(x) = \sum_{i=0}^4 \left[\binom{4}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{4-k} \right] \cdot \left[\frac{1}{\sqrt{(k/2)(2\pi)}} e^{-\frac{(x-k)^2}{2(k/2)}} \right]$$

$$= \sum_{i=0}^4 \left(\binom{4}{k} \left(\frac{1}{2}\right)^4 \right) \left(\frac{1}{\sqrt{2\pi(k/2)}} \cdot e^{-\frac{(x-k)^2}{(k/2)}} \right)$$

$\left\{ \begin{array}{l} k=0 \rightarrow X=0 \text{ (no waiting time with certainty)} \\ \rightarrow \text{Point mass} \rightarrow \text{Not normal distr at } k=0 \Rightarrow \text{PDF not Normal.} \end{array} \right.$

$$b) X = X_1 + X_2 + \dots + X_N$$

$$\hookrightarrow X_0 \Rightarrow 0 \quad (\text{no waiting time})$$

$$\hookrightarrow X_1 \dots X_4 \Rightarrow x \quad (\text{wait time}).$$

Walkthrough

$$\cancel{M_{X|N=K}(s)} = \left(e^{s + s^2/8} \right)^K$$

$$\cancel{M_{X|N}} \cdot M_Y(s) = e^{\mu s + \sigma^2 s^2/2}$$

$$M_{X|N=K}(s) = e^{(1)s + (\frac{K}{4})s^2/2} = e^{s + s^2/8}$$

$$\cancel{M_{X|N=K}(s)} = \left(e^{s + s^2/8} \right)^K$$

$$\text{Binomial}(4, 1/2) : \quad \cancel{M_N(s)} = \left(\frac{1 + e^s}{2} \right)^4$$

$$\begin{aligned} M_N(s) &= (1 - p + pe^s)^n \\ &= \left(1 - (1/2) + 1/2 e^s \right)^4 \\ &= \left(\frac{1}{2} + \frac{1}{2} e^s \right)^4 \\ &= \left(\frac{1 + e^s}{2} \right)^4 \end{aligned}$$

$$e^s \Rightarrow M_X(s)$$

$$\begin{aligned} M_N(s) &= \left[\frac{1 + \cancel{\frac{1 + e^s}{2}} (e^{s + s^2/8})}{2} \right]^4 \\ &= \left(\frac{1 + e^{s + s^2/8}}{2} \right)^4 \quad \odot \end{aligned}$$

23c) Late Dates $\rightarrow$ Success

$N \rightarrow$ number of dates until breakup.

$E[N] = ?$

~~When~~

$p \rightarrow$ Late > 45 min probability

$1-p \rightarrow$ Good date

$$E[N_n | N_{n-1}] = \cancel{P(0 \leq X \leq 0.4)} \cancel{P(0.4 \leq X \leq 1.75)}$$

$$\cancel{P(X \geq 1.75)} = ?$$

$$= P(\cancel{X} 1.75 \leq X \leq 2) = \frac{2 - 1.75}{2} = \frac{1}{8}$$

$$p \rightarrow P(\text{Bad Date}) = P(X \geq 1.75) = \frac{1}{8}$$

$$1-p \rightarrow P(\text{Good Date}) = 1 - \frac{1}{8} = \frac{7}{8}$$

~~$N_1 \rightarrow$~~

$N_1 \rightarrow$ Number dates to first bad date

geometric / binomial

$N_2 \rightarrow$ ^{Additional} Number dates to second bad date

~~$E[N_1] = E[N_2]$~~

$$N = N_1 + N_2 = 2N_1 \quad (\text{independent dates})$$

$$\cancel{N_1 = (1-p)^{k-1} \cdot p} \quad \cancel{(1 - (\frac{7}{8})^{k-1}) (\frac{1}{8})}$$

$$E[N] = 2E[N_1] = 2\left(\frac{1}{p}\right) = 2\left(\frac{1}{1/8}\right) = \underline{16} \text{ dates to breakup.}$$

1. Prob of event $P(\text{bad date}) = ?$

2. Dist to first success $\rightarrow$ Geo.

3. Note dates/events independent

$$4. \hookrightarrow E[N_2 | N_1] = 2E[N_1]$$

5. Use dist (Geometric) to get $E[N_1]$

24) Prof arrive $X = \text{uniform } [9 \text{ am} ; 1 \text{ pm}] [0, 4]$
 Duration (Task) Exponential $\lambda(y) = \frac{1}{5-y}$

$y \rightarrow$ length time 9 to X

a) $E[X] = \frac{4-0}{2} = 2 \text{ hrs (11 am)}$

$y = 2$
 $\lambda(y) = \lambda(2) = \frac{1}{5-(2)} = \frac{1}{3}$

$E[\lambda(y)] = E[\lambda(2)] = \frac{1}{\lambda} = \frac{1}{(1/3)}$
 $= 3 \text{ hours?}$

$X \rightarrow$ len task $Y \rightarrow$ length time 9am to arrive

$E[Y] = 2$ $E[X | Y=y] = \frac{1}{\lambda(y)} = 5-y$
 $\therefore E[X | Y] = 5-Y$

$E[X] = E[E[X|Y]] = E[5-Y] = 5 - E[Y] = 5-2 = 3$

b) $X \rightarrow$ arrive $Y \rightarrow$ duration

$E[X+Y] = E[X] + E[Y] = (11 \text{ am}) + (3 \text{ hours})$

$= 2 \text{ p.m.}$
 (5 hours after 9 a.m.)

c) Student uniform $[9 \text{ a.m.} ; 5 \text{ p.m.}] \rightarrow$ no prof? $\rightarrow$ leave
 $[0 ; 7]$ prof? $\rightarrow$ uniform $[0, 1]$ duration

$Z \rightarrow$ time Prof spends with student.

$K \rightarrow$ time Student arrives

$[0 \rightarrow 5 \rightarrow 7]$

$E[Z] = P(0 \leq K \leq 5) E[Z | K=k] + P(K > 5) E[Z | K=k]$
 $= \left(\frac{5}{7}\right) \left(\frac{1-0}{2}\right) + \left(\frac{2}{7}\right) (0)$
 $= \frac{5}{14} \text{ hours.}$

$E[X+Y+Z] = E[X+Y] + E[Z] = (5) + \left(\frac{5}{14}\right) = \frac{75}{14} \text{ hours after 9 a.m.}$
 5.35 hrs / 9 a.m. $\rightarrow$ 5.16 $\rightarrow$ 14:22 Prof leaves

24c) Work through

Random Variables

$W \rightarrow$ Student arrival time Uniform (9am, 5pm)

$R \rightarrow$ Student time spent with Prof Uniform (0hr, 1hr)

$T \rightarrow$ Time Prof spends with Student

$F \rightarrow$ event Student finds Prof

$X \rightarrow$ time Prof devotes to task

$Y \rightarrow$ length time betw 9am & Prof arrival

Find $E[T]$:

$$E[T] = P(F)E[T|F] + P(F^c)E[T|F^c]$$

$$= P(F)\left(\frac{1-0}{2}\right) + P(F^c)(0)$$

$$= P(F)\left(\frac{1}{2}\right)$$

$$E[T] = \frac{1}{2}P(F)$$

Find $P(F)$:

~~W between $[0$ (9am), 8 (5pm)]~~

$$P(F) = P(Y \leq W \leq X+Y)$$

$\hookrightarrow W$ between $[0$ (9am), 8 (5pm)]

$\hookrightarrow Y$ between $[0$ (9am), 4 (1pm)]

① $P(W < Y) \Rightarrow$ independent so Joint is marginal product

$$= \int_0^4 \int_0^y \left(\frac{1}{8-0}\right) \left(\frac{1}{4-0}\right) dw dy$$

$$= \int_0^4 \left(\frac{1}{4}\right) \frac{1}{4} dy = \int_0^4 \left(\frac{1}{4}\right) \left(\frac{1}{4}y\right) dy$$

$$= \frac{1}{32} \left(\frac{1}{2}\right) y^2 \Big|_0^4 = \frac{1}{64} (4)^2 - 0 = \frac{16}{64}$$

$$= \frac{16}{32} = \frac{1}{2} = \frac{1}{4}$$

② $P(W > X+Y) =$

◦ Section 4.5 Sum Rand Num Indep RVs

41) $N \rightarrow$ number people enter elevator Poisson λ .
 $W \rightarrow$ weight of each person Uniform $[100, 200]$ lbs (independent)

$X_i \rightarrow$ fraction of 100 by which i th person exceeds 100 lbs.
 $Y \rightarrow$ sum of X_i

a) $M_Y(s) = M_N(\log M_X(s))$ $[0, 1]$

$$= M_N\left(\log\left(\frac{e^{sb} - e^{sa}}{s(b-a)}\right)\right) \quad \cancel{M_X = \frac{e^{sb} - e^{sa}}{s(b-a)}}$$

$$= \left(\frac{e^s - 1}{s}\right) \log\left(\frac{e^s - 1}{s}\right) \quad = \frac{e^s - 1}{s}$$

$$= \left(e^{\lambda(e^s - 1)}\right) \log\left(\frac{e^s - 1}{s}\right) \quad M_N = e^{\lambda(e^s - 1)}$$

~~$M_N M_Y(s) = e^{\lambda\left(\frac{e^s - 1}{s} - 1\right)}$~~ ✓

b) $M_Y(s) = e^{\lambda\left(\frac{e^s - 1}{s} - 1\right)}$

~~$\frac{d}{ds} M_Y(s) = \frac{d}{ds} E[Y]$~~

~~$s=0 : \frac{d}{ds} M_Y = \frac{d}{ds} \left(e^{\lambda\left(\frac{e^s - 1}{s} - 1\right)} \right)$~~

$$= \frac{d}{ds} \left(e^{\lambda \cdot \frac{e^s - 1}{s} - \lambda} \right)$$

$$= \left(e^{\lambda \cdot \frac{e^s - 1}{s} - \lambda} \right) \frac{d}{ds} M_Y(s)$$

$M_Y(s) = e^{g(s)}$
 $\frac{d}{ds} M_Y(s) = g'(s) \cdot e^{g(s)}$
 $e^{g(0)} = 1$

$\therefore \frac{d}{ds} M_Y(0) = g'(0) = \frac{d}{ds} \left(\frac{e^s - 1}{s} \right) = \frac{(e^s)'(s) - (e^s - 1)(1)}{s^2}$

$$= \frac{se^s - e^s + 1}{s^2}$$

~~$\frac{(s+1)e^s - 1}{s^2}$~~
 ~~$= \frac{(0)e^{(0)} - e^{(0)} + 1}{(0)^2}$~~
 ~~$= \frac{0}{0}$~~

~~$$b) E[Y] = \frac{d}{ds} M_Y(s) \Big|_{s=0}$$~~

~~$$\begin{aligned} E[Y] &= E[N]E[X] = E[X]E[Y] \\ &= \left(\frac{d}{ds} M_X(s) \Big|_{s=0} \right) \left(\lambda e^{\lambda(M_X(s)-1)} \Big|_{s=0} \right) \\ &\rightarrow \{M_X(0)=1\} \\ &= \left(\frac{e^s - 1}{s} \right)' \Big|_{s=0} \left(\dots \right) \\ &= \end{aligned}$$~~

$$b) E[Y] = \frac{d}{ds} M_Y(s) \Big|_{s=0} = \frac{d}{ds} M_X(s) \Big|_{s=0} \cdot \lambda e^{\lambda(M_X(s)-1)} \Big|_{s=0}$$

$$\{M_X(0) = 1\}$$

$$\begin{aligned} &= E[X] \cdot \lambda \cdot e^{\lambda(M_X(0)-1)} \\ &= \left(\frac{1+0}{2} \right) (\lambda) (e^0) \\ &= \frac{\lambda}{2} \end{aligned}$$

$$c) E[Y] = E[E[Y|N=n]] = E[N E[X]] = E[N]E[X]$$

$$= (\lambda) \left(\frac{1+0}{2} \right) = \frac{\lambda}{2} \quad \checkmark$$

↳ poisson ↳ uniform →

° Section 8.1

1) $P(PN) = 0.5$ ← Think / Believe
 $P(7DPN \text{ busy}) = 0.01$ ← Observe / Measure
~~P(PN)~~ $P(\text{Correct}) = ?$

→ H_0 : Number is 2537267
 H_1 : Number is not 2537267

Prior : $P(H_0) = P(H_1) = 0.5$

Model : Number was busy Event : B

$$P(B | H_0) = 1$$

$$P(B | H_1) = 0.01$$

Build model based
on observation

Posterior :

$$P_{\theta|x}(\theta | x) = \frac{P_{\theta}(\theta) \cdot P_{x|\theta}(x | \theta)}{\sum_{\theta'} P_{\theta}(\theta') \cdot P_{x|\theta}(x | \theta')}$$

$$\cancel{P(H_0 | B)} = \frac{P(H_0) P(B | H_0)}{P(H_0) P(B | H_0) + P(H_1) P(B | H_1)}$$

$$= \frac{(0.5)(1)}{(0.5)(1) + (0.5)(0.01)}$$

$$= 0.990099$$

$$= \underline{0.99}$$

Select Bayes
Rule and
Apply.

2) Nefeli $\rightarrow$ multiple choice test 10 questions
3 choices per question

For each question:

① $\frac{1}{2}$: Know answer $\rightarrow$ select it / Success

② $\frac{1}{2}$: Guess answer $\rightarrow$ Success $\frac{1}{3}$.

a) Q1 correct. $P(\text{Knew Answer}) = ?$

$H_0 \rightarrow$ knew correct answer

P_0 ~~$P(H_0)$~~ $H_1 \rightarrow$ ~~guess~~ guessed correct answer.

$$P_0: P(H_0) = P(H_1) = 0.5$$

M: Event ~~C~~ $\rightarrow$ guessed correctly correct

~~$$P(H_0)$$~~
$$P(C | H_0) = 1$$

$$P(C | H_1) = \frac{1}{3}$$

$$P_0: P_{0|x}(x|x) = \frac{P_0(\theta) \cdot P_{x|\theta}(x|\theta)}{\sum_{\theta'} P_0(\theta') \cdot P_{x|\theta'}(x|\theta')}$$

$$P(H_0 | C) = \frac{(0.5)(1)}{(0.5)(1) + (0.5)(\frac{1}{3})} = \frac{3}{4} \checkmark$$

b) ~~$P_C(x) = \frac{3}{4}$~~ $\times$ $p = \frac{3}{4}$ $n = 6$

successes out of $n = 6$ with $p = \frac{3}{4}$

$\hookrightarrow$ Binomial ($n = 6, p = \frac{3}{4}$)

Problem Sets 8.2

3) Prior: $f_{\theta}(\theta) = \begin{cases} 10\theta & , \text{ if } \theta \in [0, \frac{1}{5}] \\ 0 & , \text{ otherwise} \end{cases}$

Observation: Alvin wait 30 min $X = 30 \text{ min}$
 $f_{X|\theta}(30|\theta) = 10\theta$

Posterior: $f_{\theta|X}(\theta|x) = \frac{f_{\theta}(\theta) \cdot f_{X|\theta}(x|\theta)}{\int f_{\theta}(\theta') \cdot f_{X|\theta}(x|\theta') d\theta'}$

$$f_{\theta|X}(\theta|30) = \frac{f_{\theta}(\theta) \cdot f_{X|\theta}(30|\theta)}{\int f_{\theta}(\theta') \cdot f_{X|\theta}(30|\theta') d\theta'}$$

$$= \frac{10\theta \cdot f_{X|\theta}(30|\theta)}{\int 10\theta' \cdot f_{X|\theta}(30|\theta') d\theta'}$$

Exponentially distributed: $f_{X|\theta}(x|\theta) = \lambda e^{-\lambda x}$, $x > 0$
 $f_{30|\theta}(30|\theta) = \theta e^{-30\theta}$, $x > 0$

Posterior PDF: $f_{\theta|X}(\theta|30) = \begin{cases} \frac{10\theta(\theta e^{-30\theta})}{\int_0^{1/5} 10\theta'(\theta' e^{-30\theta'}) d\theta'} ; \text{ if } \theta \in [0, \frac{1}{5}] \\ 0 ; \text{ otherwise} \end{cases}$

MAP: $\hat{\theta}_{MAP} = \arg \max f_{\theta|X}(\theta|x)$
 $= \arg \max f_{\theta|30}(\theta|30)$
 $= \arg \max \frac{\theta^2 e^{-30\theta}}{\int_0^{1/5} (\theta')^2 e^{-30\theta'} d\theta'}$

$$\frac{d}{d\theta} (\theta^2 e^{-30\theta}) = 0 = (2\theta e^{-30\theta} - 30\theta^2 e^{-30\theta}) = (2 - 30\theta)\theta e^{-30\theta}$$

$\begin{cases} \theta = 0 \text{ or } \theta = 2/30 \\ (2/30)^2 e^{-30(2/30)} > 0 \end{cases} \rightarrow \hat{\theta}_{MAP} = 2/30$

LMS: $\hat{\theta}_{LMS} = E[\theta | X = x]$ ✓

$$\hat{\theta}_{LMS} = E[\theta | X = 30] = \int \theta \cdot f_{\theta|X}(\theta|x) d\theta$$

$$\hat{\theta}_{LMS} = \frac{\int_0^{1/\sqrt{5}} \theta \cdot \theta^3 \cdot e^{-30\theta} d\theta}{\int_0^{1/\sqrt{5}} (\theta')^2 e^{-30\theta'} d\theta'}$$

b) $X = \{30, 25, 15, 40, 20\}$ minutes.
 $\rightarrow$ assume independence

Posterior:

Observation: $f_{X|\theta}(x|\theta) = f_{X_1|\theta}(x_1|\theta) \cdot f_{X_2|\theta}(x_2|\theta) \cdots f_{X_5|\theta}(x_5|\theta)$
 $= (\theta e^{-x_1\theta})(\theta e^{-x_2\theta}) \cdots (\theta e^{-x_5\theta})$
 $= \theta^5 e^{-\{x_1+x_2+x_3+x_4+x_5\}\theta}$

Key!!

Independent

$\rightarrow$ handle

separately

and multiply

$$f_{X|\theta}(x|\theta) = \theta^5 e^{-(30+25+15+40+20)\theta}$$

$$= \theta^5 e^{-130\theta}$$

Posterior:

$$f_{\theta|X}(\theta|x) = \frac{f_{\theta}(\theta) f_{X|\theta}(x|\theta)}{\int f_{\theta}(\theta') f_{X|\theta}(x|\theta') d\theta'}$$

$$= \begin{cases} \frac{\theta^6 e^{-130\theta}}{\int_0^{1/\sqrt{5}} (\theta')^6 e^{-130\theta'} d\theta'} & ; \text{ if } 0 \leq \theta \leq \frac{1}{\sqrt{5}} \\ 0 & ; \text{ otherwise} \end{cases}$$

MAP: $\frac{d}{d\theta} (\theta^6 e^{-130\theta}) = 0 = 6\theta^5 e^{-130\theta} - 130\theta^6 e^{-130\theta}$
 $= (6 - 130\theta)\theta^5 e^{-130\theta}$
 $\therefore \theta = 0 \text{ or } \theta = \frac{6}{130}$
 $\therefore \hat{\theta}_{MAP} = \frac{6}{130}$ ✓

3b con.) LHS :

$$\hat{\theta}_{LMS} = E[\theta | X = \{30, 25, 15, 40, 20\}]$$

$$= \frac{\int_0^{1/5} \theta^7 e^{-130\theta} d\theta}{\int_0^{1/5} (\theta')^6 e^{-130\theta'} d\theta'}$$

5) $\theta \in [0, 1]$

* Memo Walkthrough *

$f_{\theta}(\theta) = 2 - 4|\frac{1}{2} - \theta|$; n independent coin tosses
Results $\rightarrow k$ heads
 $\hookrightarrow n-k$ tails

$\rightarrow \hat{\theta}_{MAP} \rightarrow$ optimize numerator of posterior PDF.
 $X = k$

$$\hookrightarrow f_{\theta}(\theta) \cdot f_{X|\theta}(X|\theta)$$

$$\hookrightarrow f_{\theta}(\theta) \cdot p_{X|\theta}(k|\theta) = \binom{n}{k} (2 - 4|\frac{1}{2} - \theta|) \theta^k (1-\theta)^{n-k}$$

$$= \binom{n}{k} (2 - 4|\frac{1}{2} - \theta|) \cdot \theta^k \cdot (1-\theta)^{n-k}$$

$$\hat{\theta}_{MAP} = \frac{d}{d\theta} \left[\binom{n}{k} (2 - 4|\frac{1}{2} - \theta|) \cdot \theta^k \cdot (1-\theta)^{n-k} \right] = 0$$

$\hookrightarrow$ result of $|\frac{1}{2} - \theta|$:

$\hookrightarrow$ Not differentiable at $\theta = \frac{1}{2}$

$\hookrightarrow$ two resulting cases :

$\hookrightarrow \theta < \frac{1}{2}$
 $\hookrightarrow \theta > \frac{1}{2}$

$\theta < \frac{1}{2}$: $2 - 4(\frac{1}{2} - \theta) = 2 - 2 + 4\theta = 4\theta$

$$\frac{d}{d\theta} \left[4 \binom{n}{k} \theta^k (1-\theta)^{n-k} \right]$$

const $\left[4 \binom{n}{k} \right] \left[\theta^{k+1} (1-\theta)^{n-k} \right]$

$$(\theta^{k+1} - \theta^{n+1})' = \dots$$